SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID -** SIH25050
- **Problem Statement Title -** Smart Traffic Management System for Urban Congestion
- **Theme -** Transportation & Logistics
- **PS Category -** Software
- **Team ID -** 6430
- **Team Name -** Caffeinated Coderz

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ROADS

Responsive Odisha Adaptive Driving Systems

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IDEA APPROACH

1. Data Collection

Data is collected in real-time via CCTV Cameras, IoT Sensors

2. Layer-1

- YOLOv11 detects vehicles
- Measures Queue length, speed, No. of vehicles
- **T_{clear} formula calculates Green Light duration**

3. Layer-2 : Prioritization Engine

- Compute **Comprehensive Priority Score**.
- Efficiency & Policy: weighted vehicle counts
- Safety Proxy : hard braking, tailgating detection
- Network Intelligence ($P_{imminent}$): incoming platoons

4. Integration & Communication

- **Adapt system parameters**
- Update dashboard / public status
- Dispatch platoon data to central API

NOVELTY & UNIQUENESS

Comprehensive Priority Score (CPS)
Multi-factor decision system, not limited to vehicle count.

Proactive Safety Module
Penalizes unsafe driving (hard braking, tailgating) in real time.

Green Wave Bonus
Anticipates and prioritizes lanes with incoming traffic from upstream signals.

ADDRESSING CORE PROBLEMS

Solves Unpredictability
Our AI forecasting and dynamic signal adjustment directly combat the chaos of traffic and the failure of rigid, "dumb" timers.

Masters Network Complexity
By creating "Green Corridors," our system overcomes the problem of siloed intersections that just push bottlenecks down the road.

Enables True Prioritization
With YOLO-based vehicle detection, our system instantly identifies and creates clear paths for ambulances.

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TECHNICAL APPROACH

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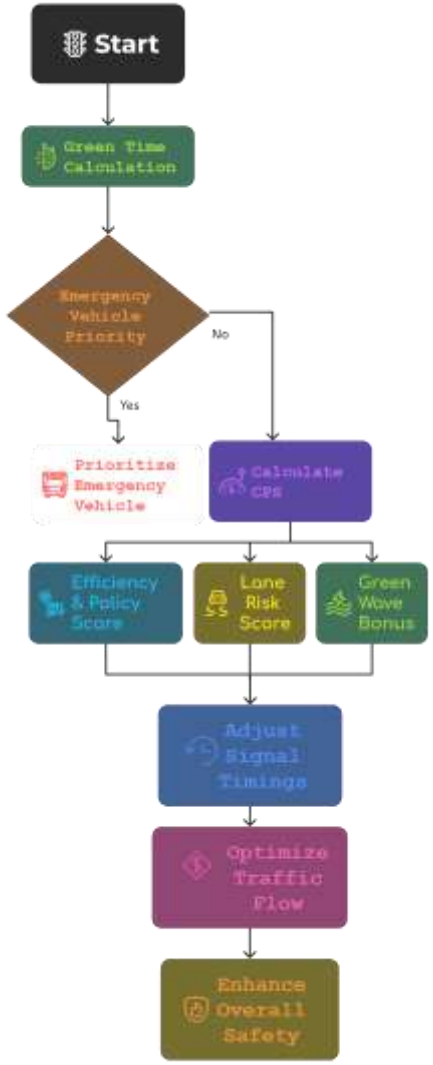
Our system operates in two Phases, *Phase-1* & *Phase-2* where Phase-2 is further divided into three parts, designed to mitigate traffic congestion with high precision & accuracy.



Phase - 1

This layer uses a formula to calculate the **optimal green signal time** required to clear vehicles at an intersection:

$$T_{clear} = 3.8s + 3.1s + 2.7s + 2.2s + (n_i - 4) \times 2.1s$$

- The **first four vehicles** take slightly longer as each starts moving in sequence (3.8s, 3.1s, 2.7s, 2.2s).
- For the **remaining vehicles**, movement stabilizes, so each is approximated at **2.1s**.

Here, $\mathbf{n_i}$ represents the total number of vehicles waiting.

This model ensures a realistic and precise estimation of the green time needed to **efficiently clear traffic queues**.

Phase – 2

The "**BRAIN**" that weighs all the factors to decide which lane wins the next green light.

The Formula:

$$\text{CPS} = \sum(w_{class} \times n_{class}) - (w_s \times C_{rate}) + (w_p \times P_{imminent})$$

Efficiency & Policy Score

Each side of an intersection is assigned points based on traffic demand and priority.

Point scheme: Ambulance = 10, Normal Vehicle = 1

Example: 2 ambulances + 10 normal vehicles → (2 × 10) + (10 × 1) = **30 points**

The lane with the higher score gets priority, ensuring both **traffic volume** and **emergency needs** are addressed.

Lane Risk Score

This module penalizes lanes where traffic behaves unsafely. Our C_{rate} (Conflict Rate) is a simple risk score derived from YOLO car detections (bounding boxes).

- Hard Braking: If a car's box rapidly shrinks between frames → +5 points
- Tailgating: If cars are too close despite free space → +2 points

These penalties may make a slightly less crowded but safer lane.

The "Green Wave" Bonus

We award bonus points when cars are about to arrive from an upstream intersection. The traffic light sends: "20 cars, ETA 30s".

The $P_{Imminent}$ score rises as ETA decrease:

- ETA = 60s → $(20 \times \left(1 - \frac{60}{60}\right)) = 0$
- ETA = 30s → $(20 \times \left(1 - \frac{30}{60}\right)) = 10$

This ensures lanes expecting incoming traffic is prioritized in advance, improving flow and efficiency.

TECH STACK

Python 3.x

OpenCV

NumPY

Flask

TensorFlow

YOLOv11

React

MongoDB

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FEASIBILITY AND VIABILITY

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Feasibility

Proven Technology:
Built on mature tech (YOLO, AI) that integrates with existing city CCTVs for immediate deployment.

Manageable Implementation
Integrates with traffic signal infra. without major hardware changes.
Minimal training required for traffic operators to interpret dashboard metrics.

Viability

Economical Viability
Reduces congestion and fuel costs, indirectly saving money for commuters.
Leverages open-source software and off-the-shelf hardware, keeping implementation costs low.

Sustainable Growth:
A self-sustaining business model fuelled by secondary data analytics revenue ensures long-term innovation

Challenges Faced by Modern Traffic Management

- **Unpredictability:** Traffic is dynamic and chaotic, requiring real-time adaptation, not rigid patterns.
- **Network Complexity:** Optimizing single intersections is useless. The city is a complex, interconnected network.
- **Emergency Prioritization:** The network must create instant, clear paths for emergency vehicles on demand.
- **Finite Supply:** Building more roads is not an option; we must manage the demand for existing space.

The Current Reality

- **Failure to Adapt:** Over 90% of signals are "dumb," using fixed timers that ignore real-time traffic and waste green light time.
- **Failure to Coordinate:** Siloed manual control and basic sensors push bottlenecks down the road.
- **Failure to Prioritize:** Unresponsive systems cannot clear paths, forcing ambulances to get stuck in traffic and lose critical minutes.
- **Failure to Manage Demand:** No data-driven tools exist to influence driver behaviour, systems only react to traffic, they don't manage it.

Proactive & Unified Solution

- **Conquering Unpredictability:** Our AI forecasts traffic 30 mins ahead, proactively adjusting signals to cut wait times by over **30%**.
- **Mastering the Network:** We create "Green Corridors" by centrally synchronizing entire arteries, reducing commuter stops by <**80%**.
- **Enabling Prioritization:** Our YOLO vision detects ambulances and instantly creates a clear path, cutting their travel time by <**50%**.

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IMPACT AND BENEFITS

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IMPACT

Adaptive Signals

Adjusts green/red light duration in real-time on the basis of lane traffic density.

Reduces congestion and optimizes overall throughput.

Emergency Priority

Gives higher priority to Ambulances, Firetrucks and other Emergency Vehicles

Ensures faster passage through intersections, potentially saving lives

Green Wave Flow

Coordinates green light for incoming upstream platoon from upstream signals

Minimizing stop-and-go driving, reducing delays and travel time.

Safety Integration

Tracks hard-breaking and tail-gating events to identify risky lanes

Reduces accident likelihood by lowering priority for unsafe lanes.

BENEFITS

Time Efficiency

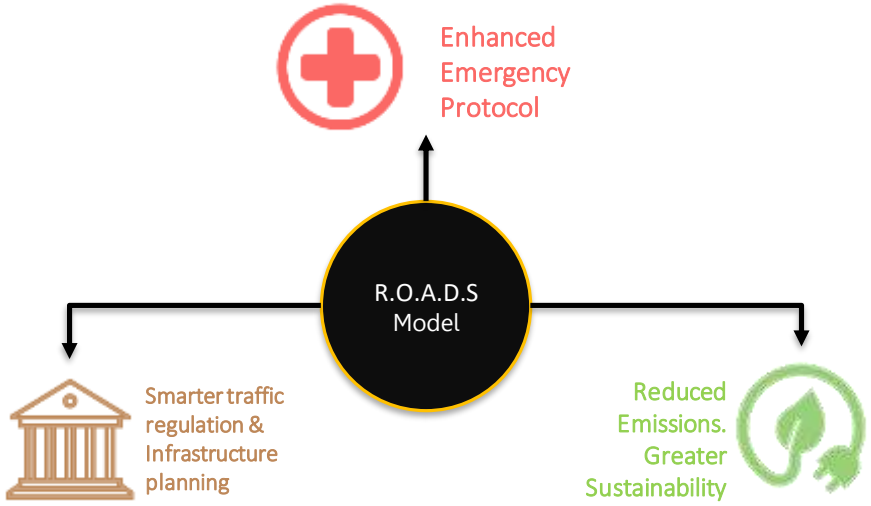
Decreases average waiting time at signals and overall commute time.
Improves traffic flow, reducing start-stop driving patterns.

Life & Safety

Decreases average waiting time at signals and overall commute time.
Improves traffic flow, reducing start-stop driving patterns.

Environmental Gains

Smooth traffic reduces fuel consumption by minimizing idling.
Lowers vehicle emissions, contributing to greener urban transport.



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RESEARCH AND REFERENCES

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- References -

[1] IDEA DESCRIPTION DETAILED

[2] R.O.A.D.S. GitHub REPOSITORY

- Research Papers -

[1] Optimization of adaptive traffic light control modes based on machine vision

[2] Self-learning adaptive traffic signal control for real-time safety optimization

- Explanation Video -